SUJAL SOTI

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EDUCATION

Institute of Engineering (IOE), Thapathali Campus, Tribhuvan University — Kathmandu, Nepal

Bachelor of Civil Engineering, First Division (74.66%) 2026

Relevant Coursework: Hydrology, Irrigation Engineering, Environmental Engineering, Fluid Mechanics, Water Supply Engineering, Engineering Geology, Numerical Methods

RESEARCH EXPERIENCE

Undergraduate Thesis: "Impacts of Climate Change on Crop Yield and Crop Water Requirement"

Supervisor: Er. Nirajan Devkota, Department of Civil Engineering, IOE Thapathali Campus

- Assessed projected climate change impacts on crop yield and irrigation water demand using AquaCrop modelling
- Performed climate projection analysis and climate data bias correction using R and Python
- Developed computational, data-driven workflows for hydro-agricultural climate impact assessment
- Key findings (Chitwan, 2021–2100): crop water requirement up to +28.5% (rice) and +43.2% (wheat); deficit irrigation recommended

Peer-Reviewed Publication (Accepted): Thapa, T., Shrestha, S., Soti, S., Gautam, V., Parajuli, S., Bhatta, S., Poudel, G.P., Chhetri, R.B., & Devkota, N. (2026). Assessment of climate change impact on crop yield and irrigation water requirement of major cereal crops in Chitwan, Nepal. H2Open Journal (Elsevier, Q2), accepted. DOI: 10.1016/j.htopen.2026.100094

PROFESSIONAL EXPERIENCE

Engineer (7th Level, Temporary – Shram Bank)
Sampada (Heritage) Department, Kathmandu Metropolitan City | Present

- Prepare quantity take-offs and cost estimates for restoration painting works on heritage structures, including temples and traditional stone water spouts (*Dhunge Dhara*).
- Coordinate with supervisors and painting teams to provide technical guidance and ensure compliance with project specifications, quality standards, and work schedules.
- Monitor restoration painting activities to support quality control, efficient resource utilization, and timely project completion.

Junior Engineer — Public Works Department, Kathmandu Metropolitan City

- Assisted in planning, supervision, and monitoring of public infrastructure projects
- Reviewed engineering drawings and technical documents; coordinated with contractors and officials

Site Engineer / Site Supervisor — Residential and Public Infrastructure Projects

- Supervised construction quality and progress on building, hydraulic, and road projects
- Prepared quantity estimates and site reports; coordinated with contractors and consultants

TECHNICAL SKILLS

Programming & Data Analysis: Python, R, Statistical Analysis, Climate Data Processing, Data Visualization, Microsoft Excel

Engineering Software: AquaCrop, AutoCAD, ETABS, SAFE, SW DTM

TEACHING & MENTORING EXPERIENCE

Peer Instructor – Survey Camp Orientation | 2024

- Delivered a 1-hour session for junior students ahead of a 10-day survey camp, covering surveying procedures, accurate data collection, field documentation, and report preparation.

RESEARCH INTERESTS

Climate Change Impact Assessment, Hydrology and Water Resources Modelling, Environmental Engineering, Sustainable Water Resources Management, Irrigation Engineering, Computational Hydrology, Drought Analysis

ACHIEVEMENTS & CERTIFICATIONS

IELTS: Overall 7.0 (Listening 8.0, Reading 6.5, Writing 6.0, Speaking 6.5)

- Registered Engineer from Nepal Engineering Council— scheduled September 2026
- Third prize in Inter model Civil Building projects, organized by CESS Thapathali.
- 2nd in Thapathali Autocad competition.